

Heart Disease Risk Assessment Report

Patient Information

Name: Notes Manger

Email: shivansh123tiwari@outlook.com

Report Date: 28/4/2025

Medical Inputs

age: 67

sex: 1

cp: 0

trestbps: 90

chol: 597

fbs: 0

restecg: 0

thalach: 78

exang: 0

oldpeak: 3

slope: 0

ca: 0

thal: 2

Prediction Results

Primary Model: Random Forest

Risk Level: Low Risk

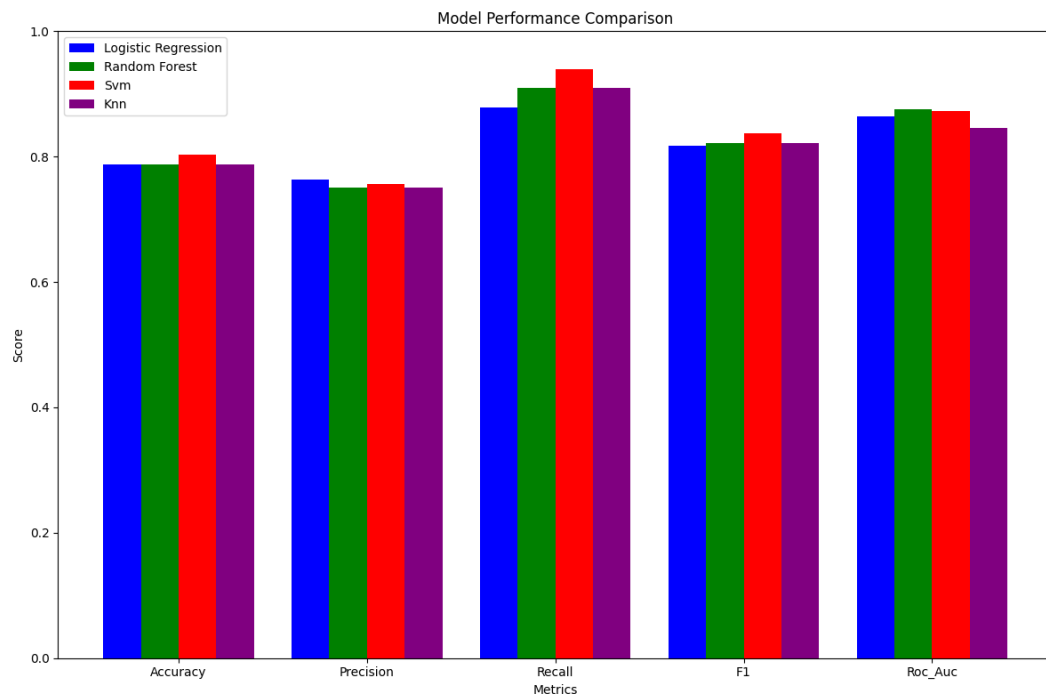
Probability: 35.00%

Model Performance Analysis

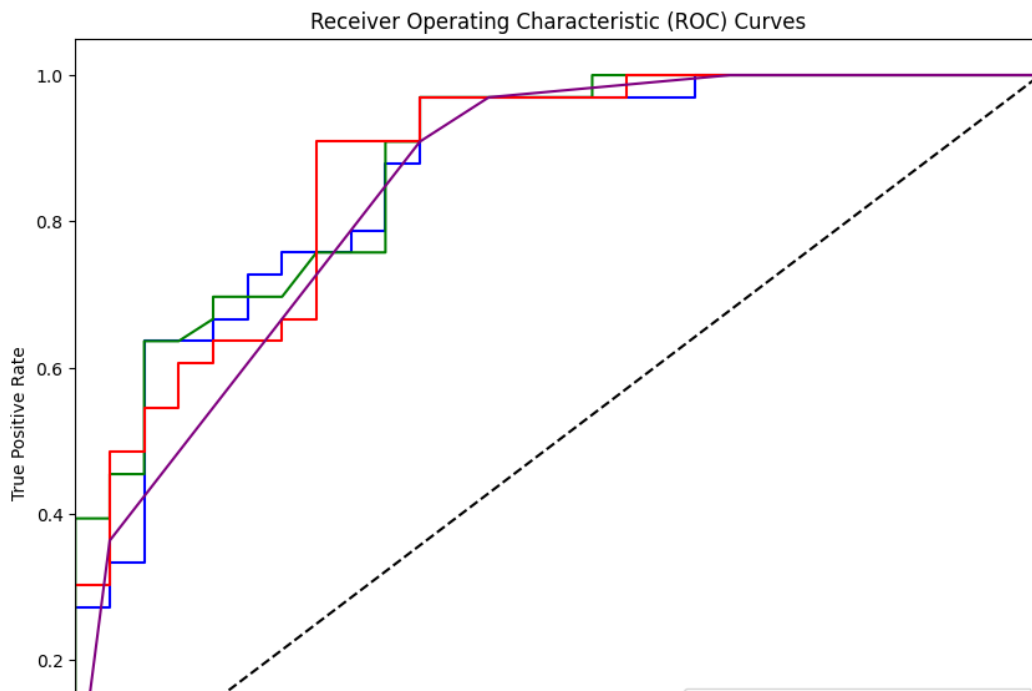
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Model Performance Notes:ð

- All models were trained on the Cleveland Heart Disease dataset (303 patients)ð
- Evaluation metrics are based on a 20% held-out test setð
- Random Forest was selected as the primary model due to highest ROC AUCð

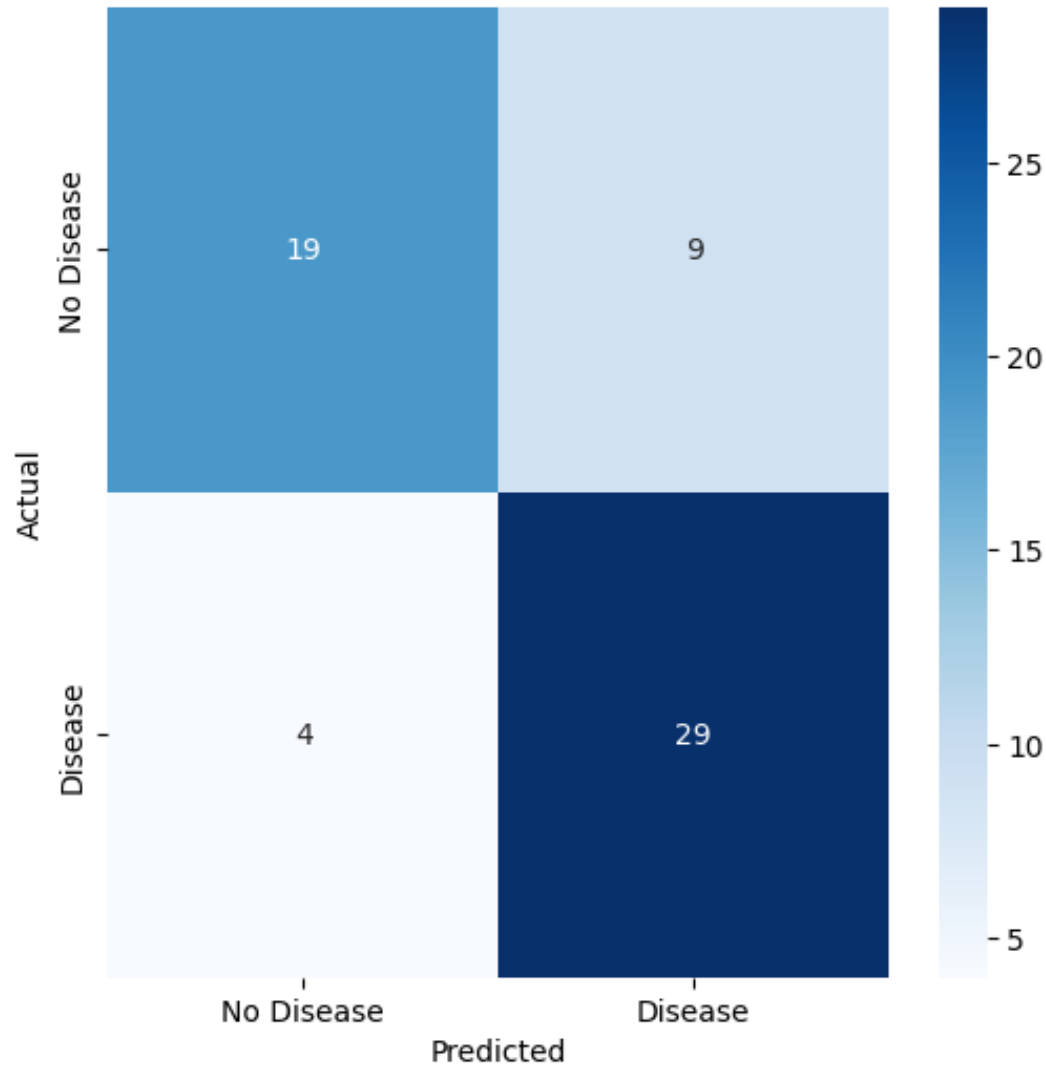


Model Performance Comparison



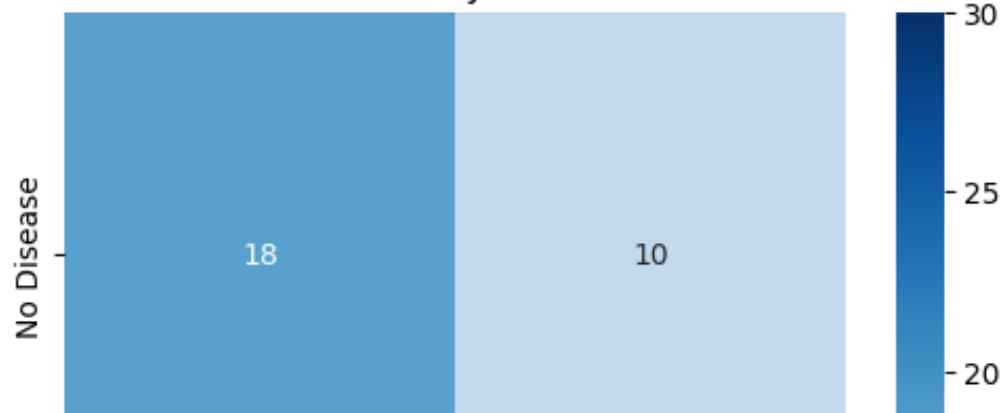
ROC Curve

Confusion Matrix - Logistic Regression
Accuracy: 0.79

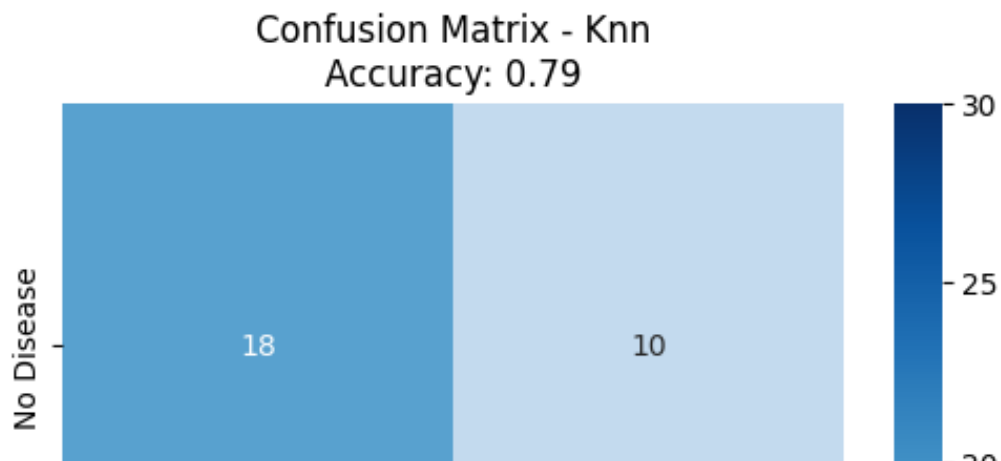
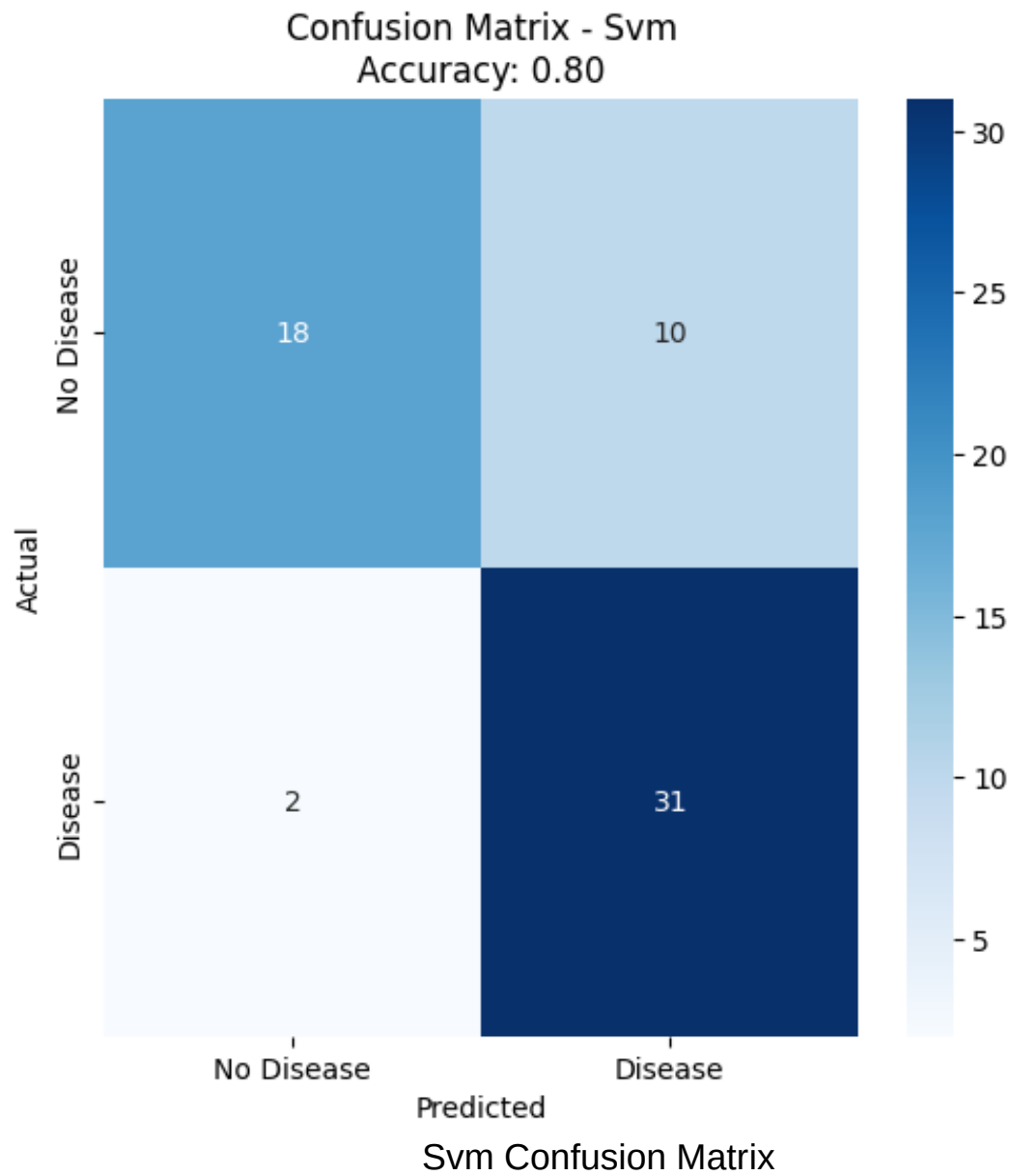


Logistic Regression Confusion Matrix

Confusion Matrix - Random Forest
Accuracy: 0.79



Random Forest Confusion Matrix



Knn Confusion Matrix

Recommendations

Based on your assessment, you appear to have low risk of heart disease.

Maintain:

- Balanced diet with omega-3 fatty acids
- 150 minutes of weekly exercise
- Healthy cholesterol levels
- Annual physical examinations
- Avoidance of tobacco products